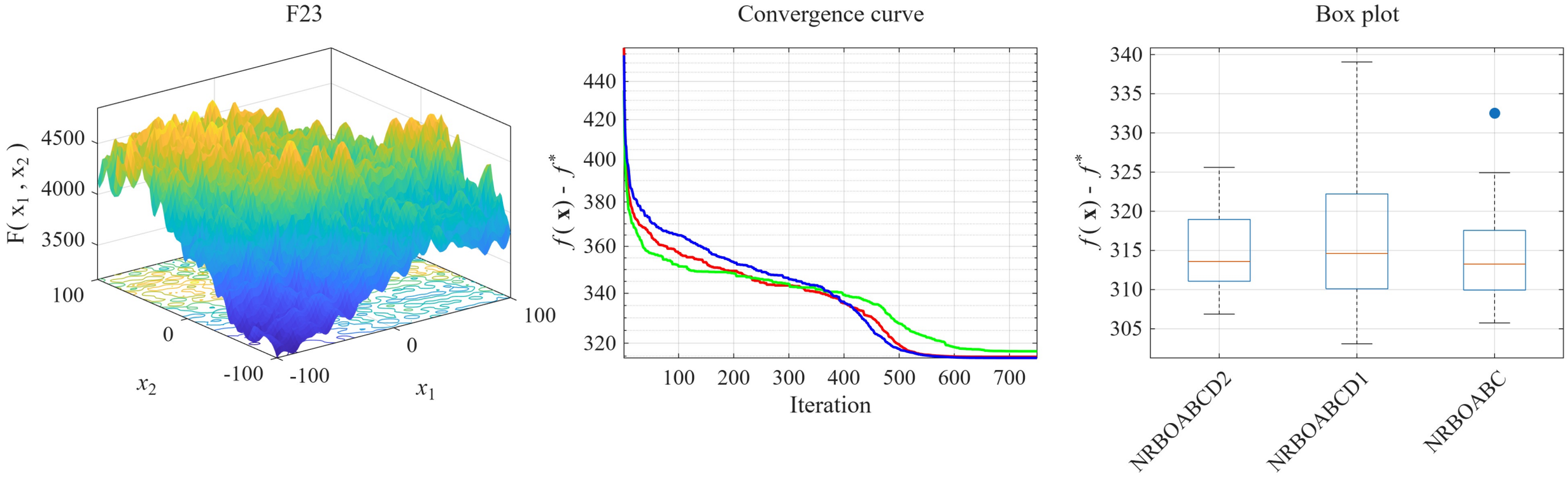

F23
F(x1 , x2)
4500
4000
3500
100
0
-100
x2
-100
0
100
x1
Convergence curve
f(x) - f*
440
420
400
380
360
340
320
100
200
300
400
500
600
700
Iteration
Box plot
f(x) - f*
340
335
330
325
320
315
310
305
NRBOABCD2
NRBOABCD1
NRBOABC
NRBOABCD2
NRBOABCD1
NRBOABC